

# 数值算法 Homework 06

姓名: 雍崔扬

学号: 21307140051

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## Problem 1

Find

$$\min_{a,b,c \in \mathbb{R}} \int_0^{\pi/2} |\sin(x) - ax^2 - bx - c|^2 dx$$

without programming.

- 换言之, 我们要求  $\sin(x)$  在  $\text{span}\{1, x, x^2\}$  上的正交投影.  
其中内积  $\langle \cdot, \cdot \rangle$  的定义为  $\langle f, g \rangle := \int_0^{\pi/2} f(x)g(x)dx$   
记  $\|\cdot\|$  为内积  $\langle \cdot, \cdot \rangle$  诱导出的范数.

**Solution:**

$$\min_{a,b,c \in \mathbb{R}} \int_0^{\pi/2} |\sin(x) - ax^2 - bx - c|^2 dx$$

$$\Updownarrow$$

$$\min_{a,b,c \in \mathbb{R}} \|\sin(x) - ax^2 - bx - c\|^2$$

$$\Updownarrow$$

$$\text{solve normal equation } \begin{bmatrix} \langle x^2, x^2 \rangle & \langle x^2, x \rangle & \langle x^2, 1 \rangle \\ \langle x, x^2 \rangle & \langle x, x \rangle & \langle x, 1 \rangle \\ \langle 1, x^2 \rangle & \langle 1, x \rangle & \langle 1, 1 \rangle \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \langle x^2, \sin(x) \rangle \\ \langle x, \sin(x) \rangle \\ \langle 1, \sin(x) \rangle \end{bmatrix}$$

注意到:

$$\begin{aligned}
\langle 1, 1 \rangle &= \int_0^{\pi/2} 1 dx = x \Big|_{x=0}^{\pi/2} = \frac{\pi}{2} \\
\langle 1, x \rangle &= \int_0^{\pi/2} x dx = \frac{1}{2} x^2 \Big|_{x=0}^{\pi/2} = \frac{\pi^2}{8} \\
\langle 1, x^2 \rangle &= \int_0^{\pi/2} x^2 dx = \frac{1}{3} x^3 \Big|_{x=0}^{\pi/2} = \frac{\pi^3}{24} \\
\langle x, x \rangle &= \int_0^{\pi/2} x^2 dx = \frac{1}{3} x^3 \Big|_{x=0}^{\pi/2} = \frac{\pi^3}{24} \\
\langle x^2, x \rangle &= \int_0^{\pi/2} x^3 dx = \frac{1}{4} x^4 \Big|_{x=0}^{\pi/2} = \frac{\pi^4}{64} \\
\langle x^2, x^2 \rangle &= \int_0^{\pi/2} x^4 dx = \frac{1}{5} x^5 \Big|_{x=0}^{\pi/2} = \frac{\pi^5}{160}
\end{aligned}

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\begin{aligned}
\langle 1, \sin(x) \rangle &= \int_0^{\pi/2} \sin(x) dx = -\cos(x) \Big|_{x=0}^{\pi/2} = 1 \\
\langle x, \sin(x) \rangle &= \int_0^{\pi/2} x \sin(x) dx \\
&= -x \cos(x) \Big|_{x=0}^{\pi/2} + \int_0^{\pi/2} \cos(x) dx \\
&= 0 + \sin(x) \Big|_{x=0}^{\pi/2} \\
&= 1 \\
\langle x^2, \sin(x) \rangle &= \int_0^{\pi/2} x^2 \sin(x) dx \\
&= -x^2 \cos(x) \Big|_{x=0}^{\pi/2} + 2 \int_0^{\pi/2} x \cos(x) dx \\
&= 0 + 2x \sin(x) \Big|_{x=0}^{\pi/2} - 2 \int_0^{\pi/2} \sin(x) dx \\
&= \pi + 2 \cos(x) \Big|_{x=0}^{\pi/2} \\
&= \pi - 2
\end{aligned}$$

因此法方程为:

$$\begin{aligned}
\begin{bmatrix} \langle x^2, x^2 \rangle & \langle x^2, x \rangle & \langle x^2, 1 \rangle \\ \langle x, x^2 \rangle & \langle x, x \rangle & \langle x, 1 \rangle \\ \langle 1, x^2 \rangle & \langle 1, x \rangle & \langle 1, 1 \rangle \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} &= \begin{bmatrix} \langle x^2, \sin(x) \rangle \\ \langle x, \sin(x) \rangle \\ \langle 1, \sin(x) \rangle \end{bmatrix} \\
&\Downarrow \\
\begin{bmatrix} \frac{\pi^5}{160} & \frac{\pi^4}{64} & \frac{\pi^3}{24} \\ \frac{\pi^4}{64} & \frac{\pi^3}{24} & \frac{\pi^2}{8} \\ \frac{\pi^3}{24} & \frac{\pi^2}{8} & \frac{\pi}{2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} &= \begin{bmatrix} \pi - 2 \\ 1 \\ 1 \end{bmatrix}
\end{aligned}$$

考虑对 Gram 矩阵进行 Gauss 消元:

$$\begin{aligned} \begin{bmatrix} \frac{\pi^5}{160} & \frac{\pi^4}{64} & \frac{\pi^3}{24} \\ \frac{\pi^4}{64} & \frac{\pi^3}{24} & \frac{\pi^2}{8} \\ \frac{\pi^3}{24} & \frac{\pi^2}{8} & \frac{\pi}{2} \end{bmatrix} &= \begin{bmatrix} 1 & & \\ \frac{5}{2\pi} & 1 & \\ \frac{20}{3\pi^2} & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\pi^5}{160} & \frac{\pi^4}{64} & \frac{\pi^3}{24} \\ 0 & \frac{\pi^3}{384} & \frac{\pi^2}{48} \\ 0 & \frac{\pi^2}{48} & \frac{2\pi}{9} \end{bmatrix} \\ &= \begin{bmatrix} 1 & & \\ \frac{5}{2\pi} & 1 & \\ \frac{20}{3\pi^2} & \frac{8}{\pi} & 1 \end{bmatrix} \begin{bmatrix} \frac{\pi^5}{160} & \frac{\pi^4}{64} & \frac{\pi^3}{24} \\ 0 & \frac{\pi^3}{384} & \frac{\pi^2}{48} \\ 0 & 0 & \frac{\pi}{18} \end{bmatrix} \end{aligned}$$

首先求解下三角方程组:

$$\begin{aligned} \begin{bmatrix} 1 & & \\ \frac{5}{2\pi} & 1 & \\ \frac{20}{3\pi^2} & \frac{8}{\pi} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} \pi - 2 \\ 1 \\ 1 \end{bmatrix} \\ \hline x &= \pi - 2 \\ y &= 1 - \frac{5}{2\pi}x = 1 - \frac{5}{2\pi}(\pi - 2) = \frac{5}{\pi} - \frac{3}{2} \\ z &= 1 - \frac{20}{3\pi^2}x - \frac{8}{\pi}y = 1 - \frac{20}{3\pi^2}(\pi - 2) - \frac{8}{\pi}\left(\frac{5}{\pi} - \frac{3}{2}\right) = 1 + \frac{16}{3\pi} - \frac{80}{3\pi^2} \end{aligned}$$

其次求解上三角方程组:

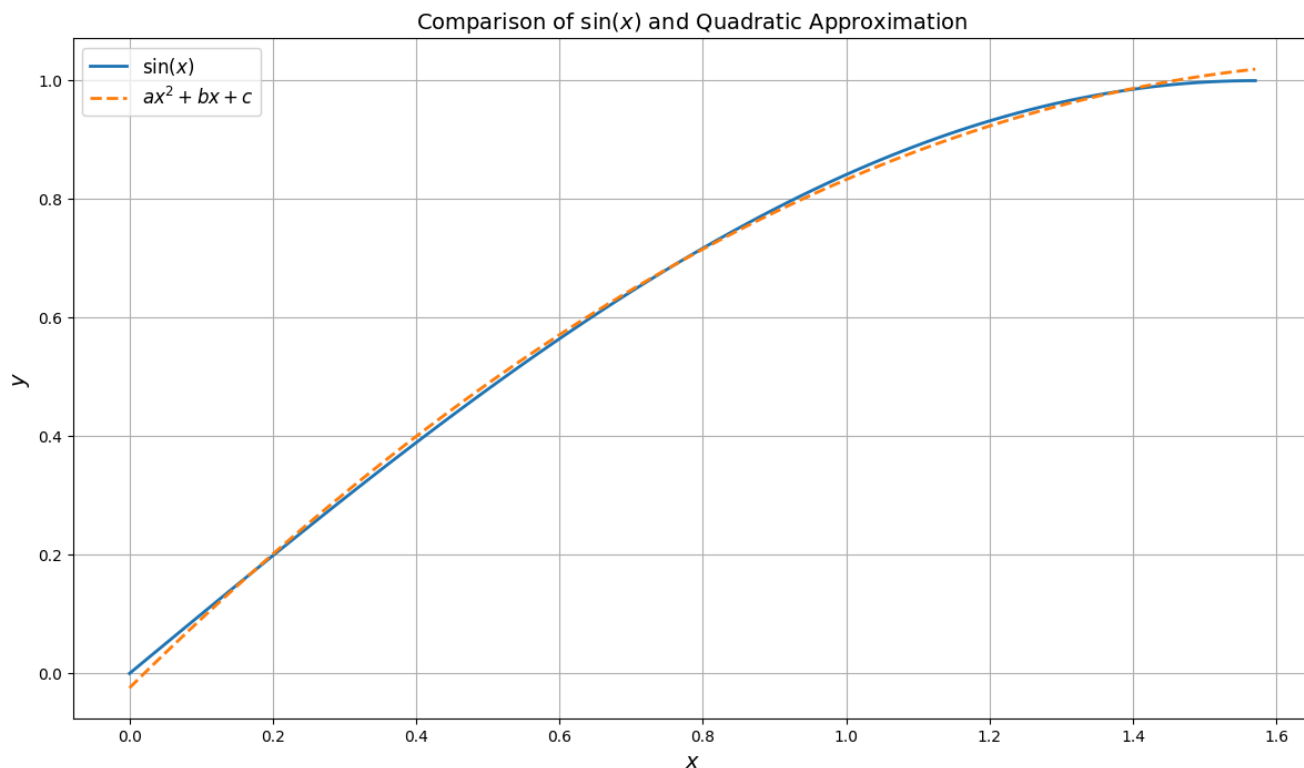
$$\begin{aligned} \begin{bmatrix} \frac{\pi^5}{160} & \frac{\pi^4}{64} & \frac{\pi^3}{24} \\ & \frac{\pi^3}{384} & \frac{\pi^2}{48} \\ & & \frac{\pi}{18} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} &= \begin{bmatrix} \pi - 2 \\ \frac{5}{\pi} - \frac{3}{2} \\ 1 + \frac{16}{3\pi} - \frac{80}{3\pi^2} \end{bmatrix} \\ \hline c_{\star} &= \frac{18}{\pi} \left( 1 + \frac{16}{3\pi} - \frac{80}{3\pi^2} \right) = \frac{6(3\pi^2 + 16\pi - 80)}{\pi^3} \\ b_{\star} &= \frac{384}{\pi^3} \left( \frac{5}{\pi} - \frac{3}{2} - \frac{\pi^2}{48} \cdot \frac{6(3\pi^2 + 16\pi - 80)}{\pi^3} \right) = \frac{48(-3\pi^2 - 28\pi + 120)}{\pi^4} \\ a_{\star} &= \frac{160}{\pi^5} \left( \pi - 2 - \frac{\pi^4}{64} \cdot \frac{48(-3\pi^2 - 28\pi + 120)}{\pi^4} - \frac{\pi^3}{24} \cdot \frac{6(3\pi^2 + 16\pi - 80)}{\pi^3} \right) = \frac{80(3\pi^2 + 36\pi - 144)}{\pi^5} \end{aligned}$$

于是问题的最优值为:

$$\begin{aligned} \min_{a,b,c \in \mathbb{R}} \|\sin(x) - ax^2 - bx - c\|^2 &= \|\sin(x)\|^2 - \|a_{\star}x^2 + b_{\star}x + c_{\star}\|^2 \\ &= \int_0^{\pi/2} (\sin(x))^2 dx - a_{\star}^2 \langle x^2, x^2 \rangle - b_{\star}^2 \langle x, x \rangle - c_{\star}^2 \langle 1, 1 \rangle - 2a_{\star}b_{\star} \langle x^2, x \rangle - 2a_{\star}c_{\star} \langle x^2, 1 \rangle - 2b_{\star}c_{\star} \langle x, 1 \rangle \\ &= \frac{1}{2} \int_0^{\pi/2} (1 - \sin(2x)) dx - a_{\star}^2 \cdot \frac{\pi^5}{160} - b_{\star}^2 \cdot \frac{\pi^3}{24} - c_{\star}^2 \cdot \frac{\pi}{2} - 2a_{\star}b_{\star} \cdot \frac{\pi^4}{64} - 2a_{\star}c_{\star} \cdot \frac{\pi^3}{24} - 2b_{\star}c_{\star} \cdot \frac{\pi^2}{8} \\ &= \frac{\pi}{4} + \frac{1}{4} \cos(2x) \Big|_{x=0}^{\pi/2} - a_{\star}^2 \cdot \frac{\pi^5}{160} - b_{\star}^2 \cdot \frac{\pi^3}{24} - c_{\star}^2 \cdot \frac{\pi}{2} - a_{\star}b_{\star} \cdot \frac{\pi^4}{32} - a_{\star}c_{\star} \cdot \frac{\pi^3}{12} - b_{\star}c_{\star} \cdot \frac{\pi^2}{4} \\ &= \frac{\pi}{4} - a_{\star}^2 \cdot \frac{\pi^5}{160} - b_{\star}^2 \cdot \frac{\pi^3}{24} - c_{\star}^2 \cdot \frac{\pi}{2} - a_{\star}b_{\star} \cdot \frac{\pi^4}{32} - a_{\star}c_{\star} \cdot \frac{\pi^3}{12} - b_{\star}c_{\star} \cdot \frac{\pi^2}{4} \\ &= \dots \\ &= -\frac{23040}{\pi^5} + \frac{11520}{\pi^4} - \frac{576}{\pi^3} - \frac{192}{\pi^2} - \frac{18}{\pi} + \frac{\pi}{4} \\ &\approx 0.00011 \end{aligned}$$

使用 `sympy` 验证的代码参见 `HW06_P1_verify.py`.

拟合结果:



## Problem 2

Find

$$\min_{a,b \in \mathbb{R}} \max_{-1 \leq x \leq 2} |x^3 + ax + b|$$

without programming.

Give an elementary proof of the correctness of your solution.

- **(偏差集 & 交错点组, 数值逼近, 定义 5.1)**

记  $C([a, b])$  为  $[a, b]$  上的连续函数空间

定义  $f \in C([a, b])$  的正偏差集、负偏差集和偏差集为:

$$\begin{aligned} E^+(f) &:= \{x \in [a, b] : f(x) = \|f\|_\infty\} \\ E^-(f) &:= \{x \in [a, b] : f(x) = -\|f\|_\infty\} \\ E(f) &:= \{x \in [a, b] : |f(x)| = \|f\|_\infty\} = E^+(f) \cup E^-(f) \end{aligned}$$

若偏差集  $E(f)$  上的点集  $\{x_1, \dots, x_k\}$  满足:

$$\begin{aligned} a \leq x_1 < x_2 < \dots < x_k \leq b \\ f(x_j) &= -f(x_{j+1}) \quad (j = 1, 2, \dots, k-1) \end{aligned}$$

则我们称  $\{x_1, \dots, x_k\}$  为  $f$  在  $[a, b]$  上的交错点组.

由  $k$  个点形成的交错点组称为是极大的, 如果不存在由  $k$  个以上点组成的交错点组.

- **(Chebyshev 定理, 数值逼近, 定理 5.4)**

记  $C([a, b])$  为  $[a, b]$  上的连续函数空间,  $\mathbb{R}_n[x]$  是至多  $n$  次的实系数多项式空间.

任意给定  $f \in C([a, b])$

若  $f \notin \mathbb{R}_n[x]$ , 则  $p$  是  $f$  的最佳一致逼近多项式 (即  $p := \arg \min_{q \in \mathbb{R}_n[x]} \|f - q\|_\infty$ ) 的充要条件是  $f - p$  在  $[a, b]$  上存在由至多  $n + 2$  个点组成的交错点组 (称为 Chebyshev 交错点组)

(值得注意的是, Chebyshev 交错点组不一定是唯一的, 点的个数也可以超过  $n + 2$ )

• (数值逼近, 推论 5.2)

如果  $f$  在  $[a, b]$  上有  $n+1$  阶导数, 且  $f^{(n+1)}$  在区间  $(a, b)$  上保号 (恒正或恒负), 则 Chebyshev 交错组唯一, 且区间  $[a, b]$  的端点属于 Chebyshev 交错组.

**Solution:**

记  $f(x) := x^3$ , 记最佳一致逼近多项式为  $p(x) := -a_*x - b_*$

根据 Chebyshev 定理可知  $f - p$  在  $[-1, 2]$  上至少有 3 个点组成的交错点组.

注意到  $f - p$  是三次函数, 因此必定有两个极值点 (记为  $x_1, x_2$ )

换言之, 方程  $3x^2 + a_* = 0$  有两个解  $x_1, x_2$ , 这表明  $a_* < 0$ , 且有  $x_{1,2} = \pm\sqrt{-a_*/3}$

同时我们知道, 无论落在  $(-1, 2)$  上的极值点是一个还是两个,

边界点  $x = -1$  和  $x = 2$  一定是交错点.

记  $\mu := \|f - p\|_\infty = \max_{-1 \leq x \leq 2} |f(x) - p(x)| \geq 0$

- ① 若  $x_1$  和  $x_2$  都落在  $(-1, 2)$  上, 则我们有:

$$\begin{cases} f(-1) - p(-1) = -1 - a_* + b_* = -\mu \\ f(x_1) - p(-\sqrt{-a_*/3}) = \frac{a_*}{3}\sqrt{-a_*/3} - a_*\sqrt{-a_*/3} + b_* = \mu \\ f(x_2) - p(\sqrt{-a_*/3}) = -\frac{a_*}{3}\sqrt{-a_*/3} + a_*\sqrt{-a_*/3} + b_* = -\mu \\ f(2) - p(2) = 8 + 2a_* + b_* = \mu \end{cases}$$

解得:

$$\begin{cases} b_* = 0 \\ a_* = -7 \\ \mu = -6 \end{cases}$$

这与  $\mu \geq 0$  的事实相矛盾.

- ② 若  $x_1$  落在  $(-1, 2)$  上, 则  $x_2 = -x_1$  必定也落在  $(-1, 2)$  上, 可归结为 ①
- ③ 若  $x_2$  落在  $(-1, 2)$  上且  $x_1$  不落在  $(-1, 2)$  上, 则我们有:

$$\begin{cases} f(-1) - p(-1) = -1 - a_* + b_* = \mu \\ f(x_2) - p(\sqrt{-a_*/3}) = -\frac{a_*}{3}\sqrt{-a_*/3} + a_*\sqrt{-a_*/3} + b_* = -\mu \\ f(2) - p(2) = 8 + 2a_* + b_* = \mu \end{cases}$$

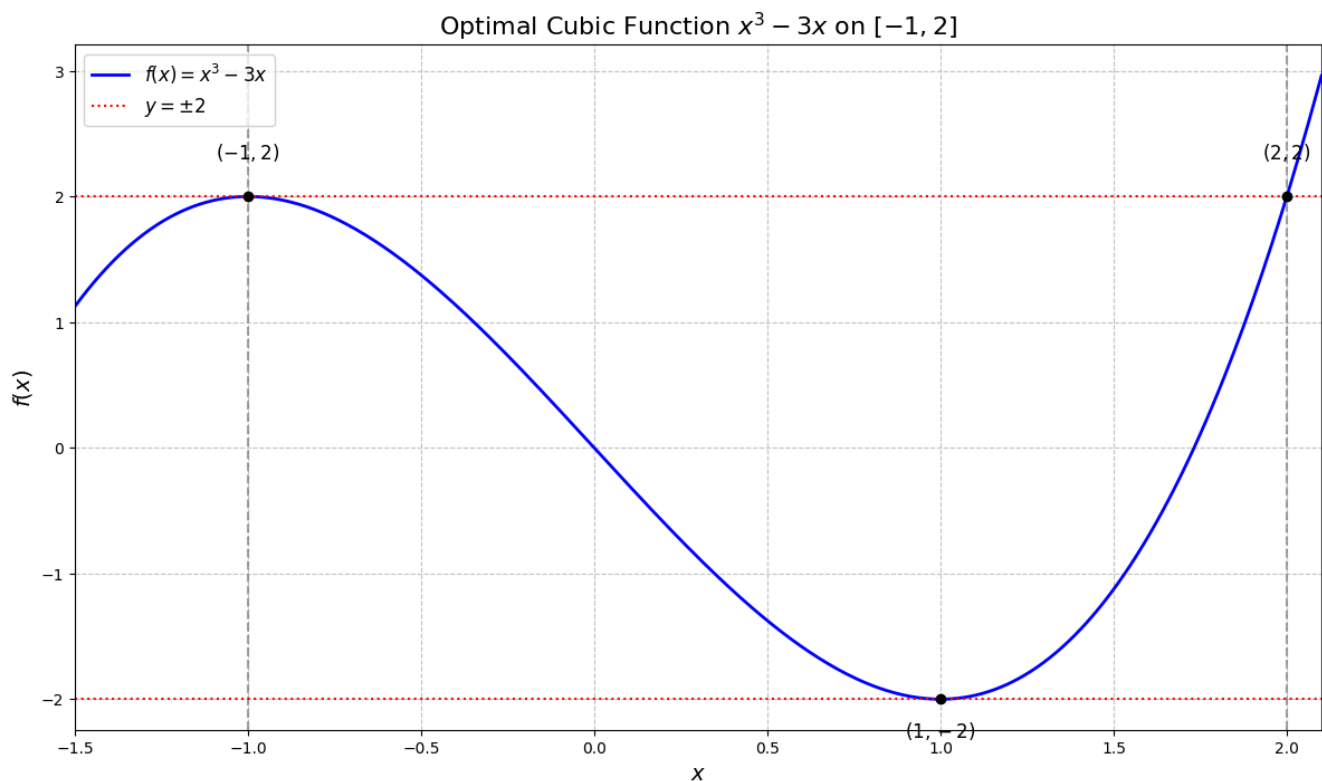
解得:

$$\begin{cases} a_* = -3 \\ b_* = 0 \\ \mu = 2 \end{cases}$$

综上所述:

$$\min_{a, b \in \mathbb{R}} \max_{-1 \leq x \leq 2} |x^3 + ax + b| = 2$$

$$\arg \min_{a, b \in \mathbb{R}} \max_{-1 \leq x \leq 2} |x^3 + ax + b| = \begin{cases} a = -3 \\ b = 0 \end{cases}$$



### 另一种观点:

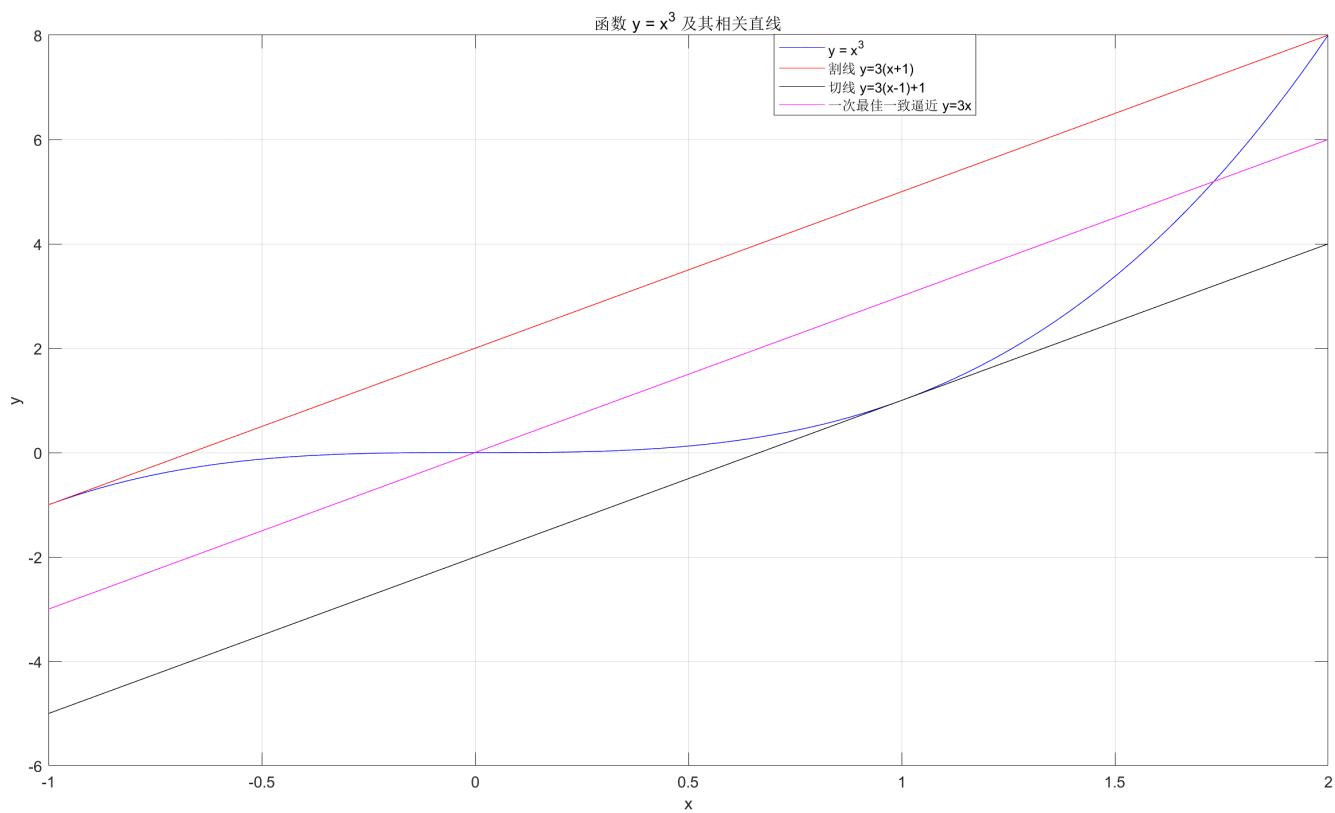
在  $[-1, 2]$  上从上方盖住  $y = x^3$ , 恰好是  $y = x^3$  在  $[-1, 2]$  上的割线  $y = 3(x + 1)$ , 碰到端点  $(-1, -1)$  和  $(2, 8)$

在  $[-1, 2]$  上从上方盖住  $y = x^3$ , 是  $y = x^3$  在  $[-1, 2]$  上的切线  $y = 3(x - 1) + 1$ , 碰到点  $(1, 1)$

取其平均就得到一次最佳一致逼近  $y = 3x$  (对应  $a = -3, b = 0$ )

容易验证,  $\{-1, 1, 2\}$  是  $y = x^3 - 3x$  在  $[-1, 2]$  上的极大 Chebyshev 交错点组.

对应的  $\|x^3 - 3x\|_\infty = 2$ .



定义残差函数  $r(x) := x^3 + ax + b$  ( $x \in [-1, 2]$ )

我们有:

$$\begin{aligned}
r(-1) &= -1 - a + b \\
r(1) &= 1 + a + b \\
r(2) &= 8 + 2a + b \\
\hline
12 &= 1 \cdot (-1 - a + b) + (-3) \cdot (1 + a + b) + 2 \cdot (8 + 2a + b) \\
&= r(-1) - 3r(1) + 2r(2) \\
&\leq |r(-1)| + 3|r(1)| + 2|r(2)| \\
&\leq 6\|r(x)\|_\infty
\end{aligned}$$

这表明对于任意  $a, b \in \mathbb{R}$  我们都有  $\|r(x)\|_\infty \geq 2$

考虑到我们已经找到了一种情况 ( $a = -3, b = 0$ ) 满足  $\|r(x)\|_\infty = 2$

因此我们有  $\min_{a, b \in \mathbb{R}} \|r(x)\|_\infty = 2$ , 即有:

$$\min_{a, b \in \mathbb{R}} \max_{-1 \leq x \leq 2} |x^3 + ax + b| = 2$$

## Problem 3

Suppose that there is a bimodal function of the form

$$y = \alpha_1 g(\beta_1(x - \gamma_1)) + \alpha_2 g(\beta_2(x - \gamma_2))$$

where  $g(x) = \exp(-x^2)$ .

Can you find out the parameters from the following noisy data set sampled from this model?

| $x_i$    | $y_i$   |
|----------|---------|
| -4.00000 | 0.00001 |
| -3.50000 | 0.00726 |
| -3.00000 | 0.25811 |
| -2.50000 | 1.87629 |
| -2.00000 | 1.55654 |
| -1.50000 | 0.17209 |
| -1.00000 | 0.00899 |
| -0.50000 | 0.05511 |
| 0.00000  | 0.24564 |
| 0.50000  | 0.60455 |
| 1.00000  | 0.89370 |
| 1.50000  | 1.03315 |
| 2.00000  | 0.51633 |
| 2.50000  | 0.18032 |
| 3.00000  | 0.04287 |
| 3.50000  | 0.00360 |
| 4.00000  | 0.00045 |

- **(optional)** Generate noisy data for high-dimensional models and try fitting the data using Gauss-Newton or Levenberg-Marquardt algorithms.
- **(optional)** Try some data fitting software (e.g. gnuplot) and compare the result produced by the software with your own solution.

## (1) 一维情形

考虑非线性最小二乘问题:

$$\min_{\beta \in \mathbb{R}^m} \|r(x, y; \beta)\|_2 := \left\| \begin{bmatrix} y_1 - \phi(x_1; \theta) \\ y_2 - \phi(x_2; \theta) \\ \vdots \\ y_n - \phi(x_n; \theta) \end{bmatrix} \right\|_2$$

Gauss-Newton 法的迭代格式为:

Given initial point  $\theta^{(0)} \in \mathbb{R}^m$  ( $m < n$ )

$k = 0$

for  $k = 0 : \text{max\_iter}$

$$J^{(k)} = \nabla_{\theta} r(x, y; \theta^{(k)}) = - \begin{bmatrix} \nabla_{\theta} \phi(x_1; \theta^{(k)})^T \\ \nabla_{\theta} \phi(x_2; \theta^{(k)})^T \\ \vdots \\ \nabla_{\theta} \phi(x_n; \theta^{(k)})^T \end{bmatrix} \in \mathbb{R}^{n \times m}$$

solve least squares problem  $\Delta \theta^{(k)} := \arg \min_{z \in \mathbb{R}^m} \|r(x, y; \theta^{(k)}) - J^{(k)} z\|_2 = (J^{(k)})^{\dagger} r(x, y; \theta^{(k)})$

$$\theta^{(k+1)} = \theta^{(k)} - \Delta \theta^{(k)}$$

if convergence criteria is reached

break

end

end

其想法与割线法类似, 每步都对残差向量  $r(x, y; \theta^{(k)})$  关于 Jacobi 矩阵  $J^{(k)}$  做线性最小二乘, 得到参数向量  $\theta^{(k)}$  的增量  $\Delta \theta^{(k)} := \arg \min_{z \in \mathbb{R}^m} \|r(x, y; \theta^{(k)}) - J^{(k)} z\|_2 = (J^{(k)})^{\dagger} r(x, y; \theta^{(k)})$  来更新  $\theta^{(k)}$ . 其中  $(J^{(k)})^{\dagger} \in \mathbb{R}^{m \times n}$  代表  $J^{(k)} \in \mathbb{R}^{n \times m}$  的 Moore-Penrose 伪逆.

给定数据点  $(x_1, y_1), \dots, (x_n, y_n)$

定义残差函数为:

$$\begin{aligned} r(\theta) &:= y - \phi(x; \theta) \\ &= y - [\alpha_1 \exp(-\beta_1^2(x - \gamma_1)^2) + \alpha_2 \exp(-\beta_2^2(x - \gamma_2)^2)] \\ \text{where } \theta &:= [\alpha_1, \beta_1, \gamma_1, \alpha_2, \beta_2, \gamma_2]^T \end{aligned}$$

我们有:

$$\nabla_{\theta} \phi(x; \theta) := \begin{bmatrix} \frac{\partial}{\partial \alpha_1} \phi(x; \theta) \\ \frac{\partial}{\partial \beta_1} \phi(x; \theta) \\ \frac{\partial}{\partial \gamma_1} \phi(x; \theta) \\ \frac{\partial}{\partial \alpha_2} \phi(x; \theta) \\ \frac{\partial}{\partial \beta_2} \phi(x; \theta) \\ \frac{\partial}{\partial \gamma_2} \phi(x; \theta) \end{bmatrix} = \begin{bmatrix} \exp(-\beta_1^2(x - \gamma_1)^2) \\ -2\alpha_1\beta_1(x - \gamma_1)^2 \exp(-\beta_1^2(x - \gamma_1)^2) \\ 2\alpha_1\beta_1^2(x - \gamma_1) \exp(-\beta_1^2(x - \gamma_1)^2) \\ \exp(-\beta_2^2(x - \gamma_2)^2) \\ -2\alpha_2\beta_2(x - \gamma_2)^2 \exp(-\beta_2^2(x - \gamma_2)^2) \\ 2\alpha_2\beta_2^2(x - \gamma_2) \exp(-\beta_2^2(x - \gamma_2)^2) \end{bmatrix}$$

计算双峰 Gauss 函数:



```

function y_pred = bimodal_gaussian(x, theta)
    % 计算双峰 Gauss 函数值
    alpha1 = theta(1);
    beta1 = theta(2);
    gamma1 = theta(3);
    alpha2 = theta(4);
    beta2 = theta(5);
    gamma2 = theta(6);

    y_pred = alpha1 * exp(-beta1.^2 * (x - gamma1).^2) + ...
        alpha2 * exp(-beta2.^2 * (x - gamma2).^2);
end

```

计算Jacobi 矩阵的函数:

```

function J = compute_jacobian(x, theta)
    % 计算 Jacobi 矩阵
    alpha1 = theta(1);
    beta1 = theta(2);
    gamma1 = theta(3);
    alpha2 = theta(4);
    beta2 = theta(5);
    gamma2 = theta(6);

    n = length(x);
    J = zeros(n, 6);

    % 第一峰的项
    x_gamma1 = x - gamma1;
    exp1 = exp(-beta1.^2 * x_gamma1.^2);
    J(:,1) = -exp1;
    J(:,2) = 2 * alpha1 * beta1 * x_gamma1.^2 .* exp1;
    J(:,3) = -2 * alpha1 * beta1.^2 * x_gamma1 .* exp1;

    % 第二峰的项
    x_gamma2 = x - gamma2;
    exp2 = exp(-beta2.^2 * x_gamma2.^2);
    J(:,4) = -exp2;
    J(:,5) = 2 * alpha2 * beta2 * x_gamma2.^2 .* exp2;
    J(:,6) = -2 * alpha2 * beta2.^2 * x_gamma2 .* exp2;
end

```

Gauss-Newton 法 (其中最小二乘子问题的求解使用 MATLAB 内置运算符  $\backslash$ ):

```

function [theta_opt, residuals, exit_flag] = fit_bimodal_gaussian(data, theta0, max_iter, tol)
    % Gauss-Newton法拟合双峰 Gauss 函数
    % 输入:
    %   data - nx2矩阵, 第一列是x, 第二列是y
    %   theta0 - 初始参数猜测 [alpha1, beta1, gamma1, alpha2, beta2, gamma2]
    %   max_iter - 最大迭代次数
    %   tol - 收敛容忍度
    % 输出:
    %   theta_opt - 优化后的参数
    %   residuals - 每次迭代的残差平方和

```

```
% exit_flag = 1表示收敛, 0表示达到最大迭代次数
```

```
x = data(:,1);
y = data(:,2);
theta = theta0(:); % 确保是列向量
residuals = zeros(max_iter,1);

for iter = 1:max_iter
    % 计算当前残差
    r = y - bimodal_gaussian(x, theta);
    residuals(iter) = sum(r.^2);

    % 计算 Jacobi 矩阵
    J = compute_jacobian(x, theta);

    % Gauss-Newton更新
    delta = J \ r;
    theta_new = theta - delta;

    % 检查收敛
    if norm(delta) < tol
        theta = theta_new;
        residuals = residuals(1:iter);
        exit_flag = 1;
        fprintf("第 %d 次迭代收敛\n", iter);
        break;
    end

    theta = theta_new;

    if iter == max_iter
        exit_flag = 0;
        warning('达到最大迭代次数而未收敛');
    end
end

theta_opt = theta;
end
```

函数调用:

```
data = [-4.00000, 0.00001;
        -3.50000, 0.00726;
        -3.00000, 0.25811;
        -2.50000, 1.87629;
        -2.00000, 1.55654;
        -1.50000, 0.17209;
        -1.00000, 0.00899;
        -0.50000, 0.05511;
        0.00000, 0.24564;
        0.50000, 0.60455;
        1.00000, 0.89370;
        1.50000, 1.03315;
        2.00000, 0.51633;
        2.50000, 0.18032;
```

```

        3.00000, 0.04287;
        3.50000, 0.00360;
        4.00000, 0.00045];
theta0 = [2, 2, -2.5, 1, 1, 1.5]';
max_iter = 10;
tol = 1e-3;
[theta_opt, residuals, exit_flag] = fit_bimodal_gaussian(data, theta0, max_iter, tol);

% 提取x和y
x = data(:,1);
y = data(:,2);

% 创建更密集的x值用于绘制平滑曲线
x_fine = linspace(min(x), max(x), 200)';

% 计算拟合曲线
y_fit = bimodal_gaussian(x_fine, theta_opt);
y_fit_data = bimodal_gaussian(x, theta_opt);

% 绘制拟合结果
figure;
subplot(2,1,1);
plot(x, y, 'ro', 'MarkerFaceColor', 'r', 'DisplayName', '原始数据');
hold on;
plot(x_fine, y_fit, 'b-', 'Linewidth', 2, 'DisplayName', '拟合曲线');
plot(x, y_fit_data, 'bx', 'MarkerSize', 8, 'Linewidth', 1.5, 'DisplayName', '拟合点');
legend('Location', 'NorthEast');
xlabel('x');
ylabel('y');
title(sprintf('双峰高斯函数拟合结果\n参数: \\alpha_1=%.3f, \\beta_1=%.3f, \\gamma_1=%.3f, \\alpha_2=%.3f, \\beta_2=%.3f, \\gamma_2=%.3f', ...
    theta_opt(1), theta_opt(2), theta_opt(3), theta_opt(4), theta_opt(5), theta_opt(6)));
grid on;

% 绘制残差收敛过程
subplot(2,1,2);
semilogx(1:length(residuals), residuals, 'ko-', 'MarkerFaceColor', 'k');
xlabel('迭代次数');
ylabel('残差平方和');
title('Gauss-Newton法收敛过程');
grid on;

% 显示最终参数
disp('优化后的参数:');
disp(['alpha1 = ', num2str(theta_opt(1))]);
disp(['beta1 = ', num2str(theta_opt(2))]);
disp(['gamma1 = ', num2str(theta_opt(3))]);
disp(['alpha2 = ', num2str(theta_opt(4))]);
disp(['beta2 = ', num2str(theta_opt(5))]);
disp(['gamma2 = ', num2str(theta_opt(6))]);
disp(['最终残差平方和 = ', num2str(residuals(end))]);

```

运行结果:

第 5 次迭代收敛

优化后的参数:

$\alpha_1 = 2.2477$

$\beta_1 = 2.0622$

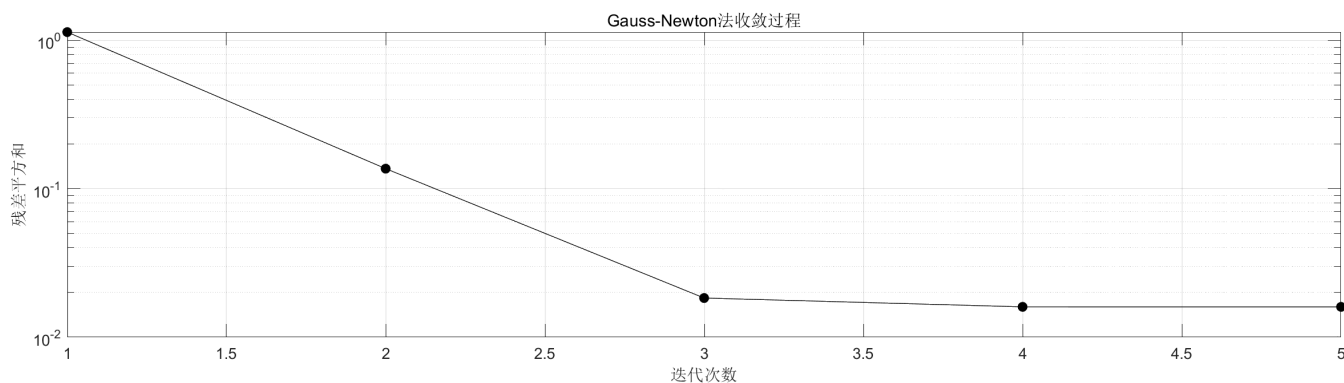
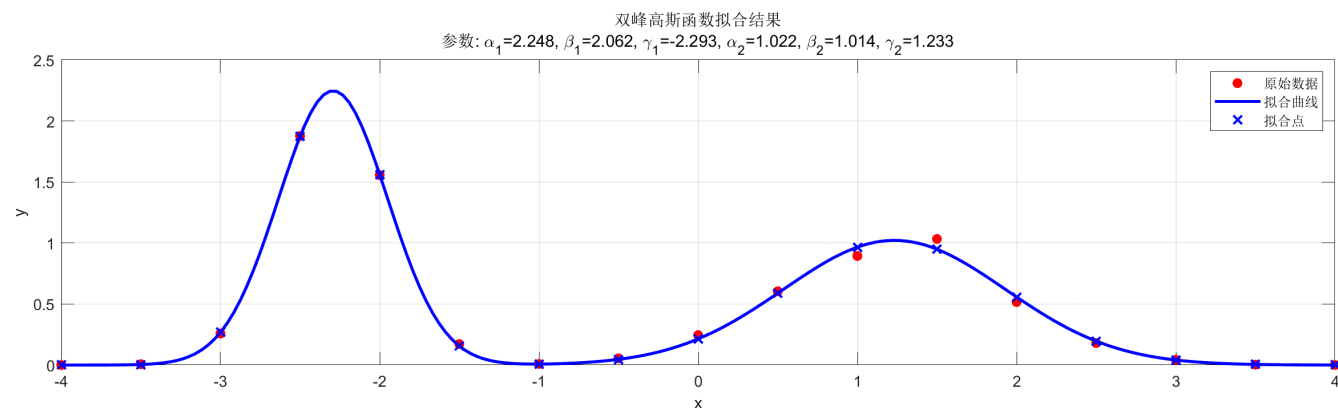
$\gamma_1 = -2.2927$

$\alpha_2 = 1.0215$

$\beta_2 = 1.0137$

$\gamma_2 = 1.2327$

最终残差平方和 = 0.016003



## Problem 4 (代码待完善)

In the homework on March 10, you have been asked to interpolate temperature in human body using a cubic spline.

In fact, interpolation is not a good idea because the data are noisy.

Try to fit the data using a periodic cubic spline with equispaced nodes  $\{1, 4, 7, \dots, 19, 22, 25\}$ .

Plot your solution for a two-day-period, and compare it with the interpolating spline.

**Solution:**

设样条  $s(x)$  在区间  $[a, b]$  上, 型值点  $(x_i, y_i)$  ( $i = 0, \dots, n$ ) 的横坐标为  $a = x_0 < x_1 < \dots < x_n = b$

假设  $s(x)$  在相邻型值点之间为三次多项式, 即  $s(x)$  在区间  $[x_i, x_{i+1}]$  上的表达式为:

$$s_i(x) := a_i + b_i x + c_i x^2 + d_i x^3 \quad (i = 0, \dots, n-1)$$

因此  $s(x)$  在整个  $[a, b]$  区间中有  $4n$  个待定系数.

它要满足以下条件:

- ① 在型值点处的函数连续: (共  $n-1$  个方程)

$$s_i(x_{i+1}) = s_{i+1}(x_{i+1}) \quad (i = 0, \dots, n-2)$$

- ② 在型值点处的一阶导数连续: (共  $n - 1$  个方程)

$$s'_i(x_{i+1}) = s'_{i+1}(x_{i+1}) \quad (i = 0, \dots, n-2)$$

- ③ 在型值点处的二阶导数连续: (共  $n - 1$  个方程)

$$s''_i(x_{i+1}) = s''_{i+1}(x_{i+1}) \quad (i = 0, \dots, n-2)$$

- ④ 周期样条 (Periodic Spline) 边界条件: (共 3 个方程)

假设样条的周期为  $b - a$ , 需要要求边界点  $(x_0 + b - a)$  和  $x_n$  处的衔接是光滑的:

$$s_0(x_0) = s_{n-1}(x_n)$$

$$s'_0(x_0) = s'_{n-1}(x_n)$$

$$s''_0(x_0) = s''_{n-1}(x_n)$$

求解  $4n$  阶线性方程组的代价  $O(n^3)$  还是太大了.

我们需要想办法将  $4n$  阶线性方程组变为低阶线性方程组.

注意到我们有  $n + 1$  个函数值  $s(x_0), \dots, s(x_n)$  和  $n + 1$  个导数值  $s'(x_0), \dots, s'(x_n)$  未知.

我们可以利用它们进行三次 Hermite 插值.

定义:

$$z_i := s(x_i) \quad (i = 0, 1, \dots, n)$$

$$k_i := s'(x_i) \quad (i = 0, 1, \dots, n)$$

$$h_i := x_{i+1} - x_i \quad (i = 0, 1, \dots, n)$$

$$\phi(x) := 3x^2 - 2x^3$$

$$\varphi(x) := x^3 - x^2$$

其中  $z_0, z_1, \dots, z_{n-1}, z_n$  和  $k_0, k_1, \dots, k_{n-1}, k_n$  都是未知的.

于是根据插值条件  $\begin{cases} s_i(x_i) = z_i \\ s_i(x_{i+1}) = z_{i+1} \\ s'_i(x_i) = k_i \\ s'_i(x_{i+1}) = k_{i+1} \end{cases}$  和三次 Hermite 插值的结论可得  $s_i(x)$  的表达式为:

$$\begin{aligned} s_i(x) &= z_i \phi\left(\frac{x_{i+1} - x}{h_i}\right) + z_{i+1} \phi\left(\frac{x - x_i}{h_i}\right) - k_i h_i \varphi\left(\frac{x_{i+1} - x}{h_i}\right) + k_{i+1} h_i \varphi\left(\frac{x - x_i}{h_i}\right) \\ s'_i(x) &= -\frac{z_i}{h_i} \phi'\left(\frac{x_{i+1} - x}{h_i}\right) + \frac{z_{i+1}}{h_i} \phi'\left(\frac{x - x_i}{h_i}\right) + k_i \varphi'\left(\frac{x_{i+1} - x}{h_i}\right) + k_{i+1} \varphi'\left(\frac{x - x_i}{h_i}\right) \\ s''_i(x) &= \frac{z_i}{h_i^2} \phi''\left(\frac{x_{i+1} - x}{h_i}\right) + \frac{z_{i+1}}{h_i^2} \phi''\left(\frac{x - x_i}{h_i}\right) - \frac{k_i}{h_i} \varphi''\left(\frac{x_{i+1} - x}{h_i}\right) + \frac{k_{i+1}}{h_i} \varphi''\left(\frac{x - x_i}{h_i}\right) \quad (i = 0, 1, \dots, n-1) \end{aligned}$$

---


$$\text{where } \phi(x) = 3x^2 - 2x^3, \varphi(x) = x^3 - x^2$$

$$\phi'(x) = 6x - 6x^2, \varphi'(x) = 3x^2 - 2x$$

$$\phi''(x) = 6 - 12x, \varphi''(x) = 6x - 2$$

现考虑二阶导的插值条件  $s''_i(x_{i+1}) = s''_{i+1}(x_{i+1})$ :

$$\begin{aligned}
s''_i(x_{i+1}) &= \frac{z_i}{h_i^2} \phi''(0) + \frac{z_{i+1}}{h_i^2} \phi''(1) - \frac{k_i}{h_i} \phi''(0) + \frac{k_{i+1}}{h_i} \phi''(1) \\
&= 6 \frac{z_i}{h_i^2} - 6 \frac{z_{i+1}}{h_i^2} + 2 \frac{k_i}{h_i} + 4 \frac{k_{i+1}}{h_i} \\
s''_{i+1}(x_{i+1}) &= \frac{z_{i+1}}{h_{i+1}^2} \phi''(1) + \frac{z_{i+2}}{h_{i+1}^2} \phi''(0) - \frac{k_{i+1}}{h_{i+1}} \phi''(1) + \frac{k_{i+2}}{h_{i+1}} \phi''(0) \\
&= -6 \frac{z_{i+1}}{h_{i+1}^2} + 6 \frac{z_{i+2}}{h_{i+1}^2} - 4 \frac{k_{i+1}}{h_{i+1}} - 2 \frac{k_{i+2}}{h_{i+1}} \\
\hline
s''_i(x_{i+1}) &= s''_{i+1}(x_{i+1}) \\
&\Updownarrow \\
6 \frac{z_i}{h_i^2} - 6 \frac{z_{i+1}}{h_i^2} + 2 \frac{k_i}{h_i} + 4 \frac{k_{i+1}}{h_i} &= -6 \frac{z_{i+1}}{h_{i+1}^2} + 6 \frac{z_{i+2}}{h_{i+1}^2} - 4 \frac{k_{i+1}}{h_{i+1}} - 2 \frac{k_{i+2}}{h_{i+1}} \\
&\Updownarrow \\
\beta_i k_i + \alpha_{i+1} k_{i+1} + \beta_{i+1} k_{i+2} &= \gamma_i z_i + \eta_{i+1} z_{i+1} + \gamma_{i+1} z_{i+2} \\
\text{where } \begin{cases} h_i = x_{i+1} - x_i & (i = 0, \dots, n-1) \\ \beta_i = \frac{1}{h_i} & (i = 0, \dots, n-1) \\ \alpha_{i+1} = 2(\beta_i + \beta_{i+1}) & (i = 0, \dots, n-2) \\ \gamma_i = 3\beta_i^2 & (i = 0, \dots, n-1) \\ \eta_{i+1} = \gamma_i - \gamma_{i+1} & (i = 0, \dots, n-2) \end{cases}
\end{aligned}$$

**现考虑周期样条边界条件:**

假设样条的周期为  $b - a = x_n - x_0$ , 要求边界点  $x_0$  和  $x_n$  处的衔接是光滑的:

$$\begin{aligned}
s_0(x_0) &= s_{n-1}(x_n) \\
s'_0(x_0) &= s'_{n-1}(x_n) \\
s''_0(x_0) &= s''_{n-1}(x_n) \\
&\Updownarrow \\
z_0 = s_0(x_0) &= s_{n-1}(x_n) = z_n \\
k_0 = s'_0(x_0) &= s'_{n-1}(x_n) = k_n \\
2(-\gamma_0 z_0 + \gamma_0 z_1 - 2\beta_0 k_0 - \beta_0 k_1) &= s''_0(x_0) = s''_{n-1}(x_n) = 2(\gamma_{n-1} z_{n-1} - \gamma_{n-1} z_n + \beta_{n-1} k_{n-1} + 2\beta_{n-1} k_n) \\
&\Downarrow \\
2(\beta_0 + \beta_{n-1})k_0 + \beta_0 k_1 + \beta_{n-1} k_{n-1} &= \gamma_0 z_0 - \gamma_0 z_1 + \gamma_{n-1} z_{n-1} - \gamma_{n-1} z_n
\end{aligned}$$

若额外定义  $\alpha_0 = 2(\beta_0 + \beta_{n-1})$  和  $\eta_0 = (\gamma_0 - \gamma_{n-1})$ , 并代入  $z_n = z_0$ ,

则我们有  $\alpha_0 k_0 + \beta_0 k_1 + \beta_{n-1} k_{n-1} = \eta_0 z_0 - \gamma_0 z_1 + \gamma_{n-1} z_{n-1}$

此外, 在  $\beta_{n-2} k_{n-2} + \alpha_{n-1} k_{n-1} + \beta_{n-1} k_n = \gamma_{n-2} z_{n-2} + \eta_{n-1} z_{n-1} + \gamma_{n-1} z_n$  中代入  $z_n = z_0$  和  $k_n = k_0$

即可得到  $\beta_{n-1} k_0 + \beta_{n-2} k_{n-2} + \alpha_{n-1} k_{n-1} = \gamma_{n-2} z_{n-2} + \eta_{n-1} z_{n-1} + \gamma_{n-1} z_0$

于是方程组为:

$$\begin{cases} \alpha_0 k_0 + \beta_0 k_1 + \beta_{n-1} k_{n-1} = \eta_0 z_0 - \gamma_0 z_1 + \gamma_{n-1} z_{n-1} & \text{(additional condition)} \\ \beta_i k_i + \alpha_{i+1} k_{i+1} + \beta_{i+1} k_{i+2} = \gamma_i z_i + \eta_{i+1} z_{i+1} + \gamma_{i+1} z_{i+2} & (i = 0, \dots, n-3) \\ \beta_{n-1} k_0 + \beta_{n-2} k_{n-2} + \alpha_{n-1} k_{n-1} = \gamma_{n-2} z_{n-2} + \eta_{n-1} z_{n-1} + \gamma_{n-1} z_0 & \text{(additional condition)} \end{cases}$$

$$\Updownarrow$$

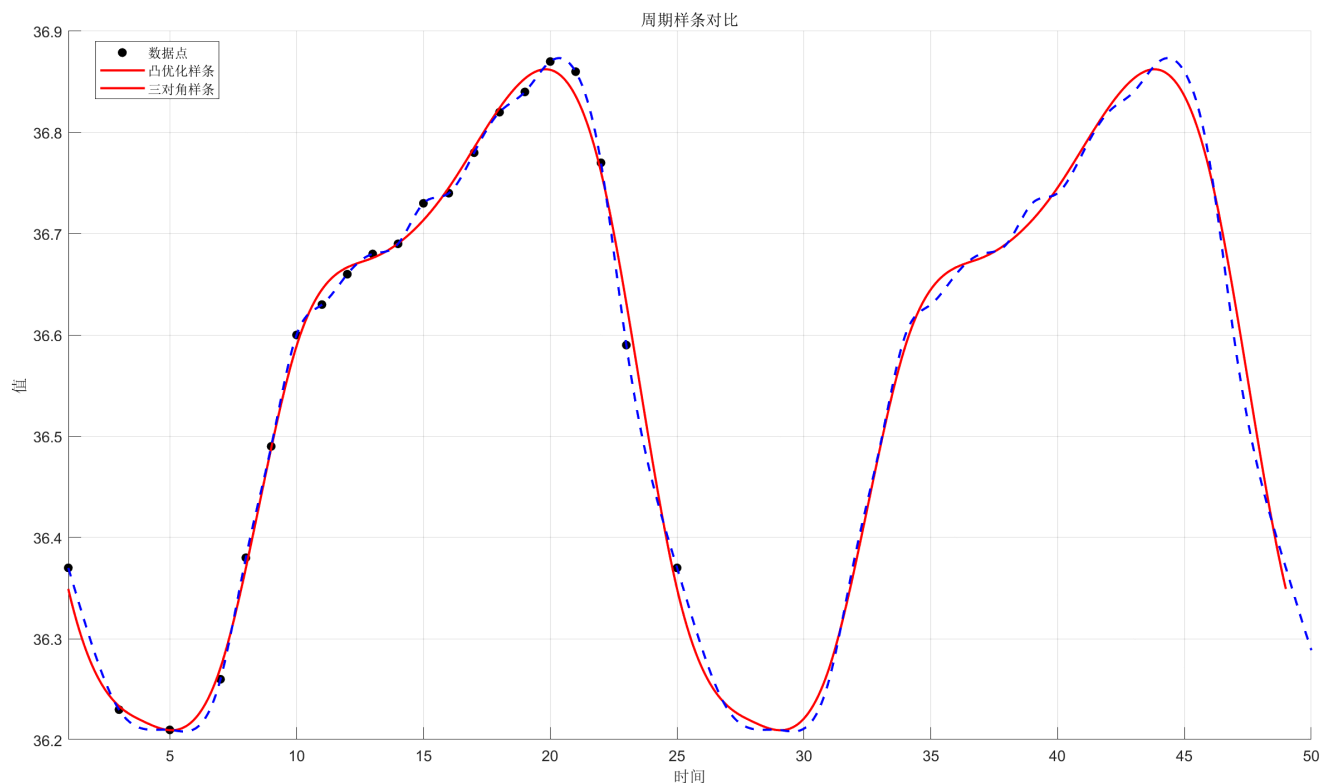
$$\left[ \begin{array}{cccc|cccc} \alpha_0 & \beta_0 & & & \beta_{n-1} & -\eta_0 & \gamma_0 & & & -\gamma_{n-1} \\ \beta_0 & \alpha_1 & \beta_1 & & & -\gamma_0 & -\eta_1 & -\gamma_1 & & \\ & \beta_1 & \alpha_2 & \beta_2 & & & -\gamma_1 & -\eta_2 & -\gamma_2 & \\ & & & \ddots & \ddots & \ddots & & \ddots & \ddots & \\ & & & & \beta_{n-3} & \alpha_{n-2} & \beta_{n-2} & & & \\ \beta_{n-1} & & & & \beta_{n-2} & \alpha_{n-1} & & -\gamma_{n-1} & & \end{array} \right] \begin{bmatrix} k_0 \\ k_1 \\ k_2 \\ \vdots \\ k_{n-2} \\ k_{n-1} \\ z_0 \\ z_1 \\ z_2 \\ \vdots \\ z_{n-2} \\ z_{n-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}$$

$$\Updownarrow$$

$$Aq = 0_n$$

where  $A \in \mathbb{R}^{n \times 2n}$  and  $q \in \mathbb{R}^{2n}$

我们可以将  $Aq = 0_n$  作为线性约束，最小化在数据点上的平方误差。  
运行结果:



## Problem 5

Write a program to compute the roots of a few low-degree Legendre polynomials.

- **罗一凡:** 递归二分法, 预估  $O(n^3)$  代价.

利用的是正交多项式的根的严格交错性质.

要计算  $n$  次正交多项式的根, 只需利用  $n - 1$  次正交多项式的根和端点  $-1, 1$  作为初始点做  $n$  个二分法.

双精度浮点数下每个二分法的迭代次数至多是 52 次,

但正交多项式计算代价是  $O(n)$  (使用秦九韶算法), 因此每次二分法的代价为  $O(n^3)$ .

我们要依次计算  $1, 2, \dots, n$  次正交多项式的根,

因此代价为  $O(1^2 + 2^2 + \dots + n^2) = O(n^3)$

#### Solution:

定义自伴线性算子  $F[p](x) := x \cdot p(x) (\forall x \in \mathbb{R})$

生成正交多项式基的过程是一个 Lanczos 过程:

```

Given polynomial function  $p_0(x)$  of degree zero such that  $\|p_0\| = 1$ 
 $k = -1; \beta_{-1} = 1; p_{-1} \equiv 0; r_{-1} = p_0$ 
while  $\beta_k \neq 0$ 
     $p_{k+1} = \frac{r_k}{\beta_k}$ 
     $k = k + 1$ 
     $\alpha_k = \langle F[p_k], p_k \rangle$ 
     $r_k = F[p_k] - \alpha_k p_k - \beta_{k-1} p_{k-1}$ 
     $\beta_k = \|r_k\|$ 
end
  
```

其中  $\{p_k\}$  和  $\{r_k\}$  均是多项式序列,  $\langle \cdot, \cdot \rangle$  和  $\|\cdot\|$  是函数空间上的内积和范数.

于是对于任意  $k \in \mathbb{Z}_+$  我们都有:

$$\begin{aligned}
 F[P_k] &= F[p_0, p_1, \dots, p_k] \\
 &= [p_0, p_1, \dots, p_k, p_{k+1}] \begin{bmatrix} \alpha_0 & \beta_0 & & & \\ \beta_0 & \alpha_1 & & & \\ & & \ddots & & \\ & & & \ddots & \beta_{k-1} \\ & & & \beta_{k-1} & \alpha_k \\ & & & & \beta_k \end{bmatrix}_{(k+2) \times (k+1)} \\
 &= P_{k+1} \tilde{T}_{k+1} \\
 &= [P_k, p_{k+1}] \begin{bmatrix} T_{k+1} \\ \beta_k e_{k+1}^T \end{bmatrix} \\
 &= P_k T_{k+1} + \beta_k p_{k+1} \\
 &= P_k T_{k+1} + r_k
 \end{aligned}$$

其中  $e_{k+1}$  是  $\mathbb{R}^{k+1}$  空间的第  $k + 1$  个标准正交基向量.

这样我们就得到了三项递推关系:

$$\begin{aligned}
 F[p_k] &= \beta_{k-1} p_{k-1} + \alpha_k p_k + \beta_k p_{k+1} \\
 \text{where } F[p_k](x) &= x \cdot p_k(x)
 \end{aligned}$$

可以证明  $p_n(x)$  有  $n$  个不同的根, 且它们是不可约对称三对角阵  $T_{n-1}$  的特征值.

Legendre 多项式是一类在区间  $[-1, 1]$  上关于权重函数 1 正交的多项式序列:

(注意这里的  $P_n$  记号和上面 Lanczos 过程中的  $P_n$  记号重复了, 它们指代的不是同一种东西)



$$P_n(x) := \frac{((x^2 - 1)^n)^{(n)}}{2^n n!}$$

$$\int_{-1}^1 P_m(x) P_n(x) dx = \begin{cases} 0, & \text{if } m \neq n \\ \frac{2}{2n+1}, & \text{if } m = n \end{cases}$$

三项递推公式:

$$\begin{aligned} P_0(x) &\equiv 1 \\ P_1(x) &= x \\ P_2(x) &= \frac{3x^2 - 1}{2} \\ P_3(x) &= \frac{5x^3 - 3x}{2} \\ \hline (n+1)P_{n+1}(x) &= (2n+1)xP_n(x) - nP_{n-1}(x) \end{aligned}$$

我们定义单位化的 Legendre 多项式为:

$$p_n(x) := \sqrt{\frac{2n+1}{2}} P_n(x)$$

对应的三项递推公式为:

$$\begin{aligned} (n+1)P_{n+1}(x) &= (2n+1)xP_n(x) - nP_{n-1}(x) \\ &\Downarrow \\ (n+1)\sqrt{\frac{2n+3}{2}}p_{n+1}(x) &= (2n+1)\sqrt{\frac{2n+1}{2}}xp_n(x) - n\sqrt{\frac{2n-1}{2}}p_{n-1}(x) \\ &\Downarrow \\ xp_n(x) &= \frac{n+1}{\sqrt{(2n+1)(2n+3)}}p_{n+1}(x) + \frac{n}{\sqrt{(2n-1)(2n+1)}}p_{n-1}(x) \end{aligned}$$

下面推导  $\alpha_k$  和  $\beta_k$  的通项公式: (其中  $k = 0, 1, \dots$ )

- ① 注意到单位化的 Legendre 多项式满足奇偶性:  $p_k(-x) = (-1)^k p_k(x)$

考虑  $\alpha_k$  的计算:

$$\begin{aligned} \alpha_k &:= \langle F[p_k], p_k \rangle \\ &= \int_{-1}^1 xp_k(x)p_k(x)dx \\ &= \int_{-1}^1 x[p_k(x)]^2 dx \quad (\text{note that } x[p_k(x)]^2 \text{ is an odd function for any } k) \\ &= 0 \end{aligned}$$

- ② 单位化的 Legendre 多项式的递推关系是: [\(reference\)](#)

$$xp_k(x) = \frac{k+1}{\sqrt{(2k+1)(2k+3)}}p_{k+1}(x) + \frac{k}{\sqrt{(2k-1)(2k+1)}}p_{k-1}(x)$$

与 Lanczos 过程的三项递推公式  $xp_k(x) = \beta_{k-1}p_{k-1}(x) + \alpha_k p_k(x) + \beta_k p_{k+1}(x)$  比较,  
(注意根据 ① 我们知道  $\alpha_k = 0$ ) 可得:

$$\beta_k = \frac{k+1}{\sqrt{(2k+1)(2k+3)}}$$

这样我们就可以构造不可约对称三对角阵  $T_{k-1} \in \mathbb{R}^{k \times k}$  了,  
其特征根即为单位化 Legendre 多项式的根, 也即 Legendre 多项式的根.  
Matlab 实现:

```
function roots = legendre_roots(n)
% 计算 n 次 Legendre 多项式的根
if n == 0
    roots = [];
    return;
elseif n == 1
    roots = 0;
    return;
end

% 构造次对角线元素 beta_k
beta = zeros(n-1, 1);
for k = 0:n-2
    beta(k+1) = (k + 1) / sqrt((2*k + 1) * (2*k + 3));
end

% 构造对称三对角矩阵 T
T = diag(beta, 1) + diag(beta, -1);

% 计算特征值并排序
roots = eig(T);
roots = sort(roots);
end
```

函数调用:

```
% 计算 3 次 Legendre 多项式的根
n = 3;
roots = legendre_roots(n);
disp(roots);
```

运行结果:

```
-0.7746
0.0000
0.7746
```

这与 3 次 Legendre 多项式  $P_3(x) = \frac{5x^3-3x}{2}$  的根  $0, \pm\sqrt{3/5}$  是吻合的.

## Problem 6 (optional)

Use a few points to interpolate the bivariate function  $z = \exp(-x^2 - y^2)$  over  $[-1, 1] \times [-1, 1]$ . Compare the bad approach using complex polynomial interpolation (see Homework 3 Problem 5) with other approaches discussed last week (e.g. Shepard interpolation)

## Problem 7 (optional)

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Hermite polynomials can be defined through one of the following ways:

- ①  $H_n(x) = (-1)^n e^{x^2} (e^{-x^2})^{(n)}$
- ②  $H_0(x) \equiv 1, H_1 = 2x, H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x) \ (n \geq 1)$
- ③  $\sum_{n=0}^{\infty} H_n(x) t^n / n! = e^{2xt - t^2}$

Show that these definitions are equivalent.

## Problem 8 (optional)

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